

- Document Control
- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
- Date: 2024-06-12
- Status: Draft
- Executive Summary
 - The Application facilitates an automated ETL pipeline involving source CSV data ingestion, transformation, cleaning, aggregation, and SCD Type 2 history management, implemented using Delta Live Tables.
- Business Objectives
 - Automate data ingestion from CSV files to Delta tables
 - Ensure data quality through null and duplicate removal
 - Aggregate customer spend and maintain accurate, auditable transaction history
 - Enable change data capture via SCD Type 2 implementation
- In-Scope
 - Source CSV data reading and delta table loading
 - Schema enforcement for customer and order tables
 - Data transformation and cleaning logic
 - Creation of ordersummary and customeraggregatespend tables
 - Aggregation logic on ordersummary data
 - SCD Type 2 implementation for ordersummary
 - Process deployment using Delta Live Tables
- Out-of-Scope
 - Any manual data manipulation
 - External system integrations beyond defined delta tables
 - Real-time or streaming ingestion
 - Data visualization or reporting dashboards
- Stakeholders
 - Data Engineering Team
 - Business Analysts
 - Compliance Team
- Business Requirements

BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table ($\text{PricePerUnit} * \text{Qty}$)
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen_ai_poc_databrickscoe, schema=sdlc_wizard if not exists
- Business Value: Centralized storage for enriched order data
- Priority: High

BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High
- Assumptions & Constraints
 - Source data is available and accessible via defined CSV file paths
 - Only the specified schemas will be enforced
 - Delta Lake infrastructure is operational
- Success Metrics
 - Successful, automated runs from CSV to target tables
 - Zero null/duplicate records post-processing
 - Accurate aggregation results in customeraggregatespend
 - SCD Type 2 audit trails present
- Risks & Mitigations
 - Source data format changes — Mitigation: Schema enforcement and validation
 - Data quality issues from source — Mitigation: Robust null and duplicate handling logic
 - Infrastructure/permission errors — Mitigation: Pre-execution checks and monitoring

- Approval
 - Business Owner: Data Engineering Lead
 - Data Engineering Lead: Designated Engineer
 - Compliance Review: Compliance Team
 - Date: 2024-06-12
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- Appendix

Appendix A: Source Definitions

Source Name | Type | Location | Format | Notes

customerdata | CSV | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata | CSV | Customer data from source system

orderdata | CSV | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata | CSV | Order data from source system

Appendix B: Target Definitions

Target Name | Layer | Table | Columns | Notes

ordersummary | Delta | sdlc_wizard.ordersummary | OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount | SCD Type 2 applied

customeraggregatespend | Delta | sdlc_wizard.customeraggregatespend | Name, TotalAmount, Date | Aggregated spend by customer

Appendix C: Transformations

Step | Transformation | Input | Output | Logic

1 | Schema enforcement | customerdata | customerdata | Map to CustId, Name, EmailId, Region

2 | Schema enforcement | orderdata | orderdata | Map to OrderId, ItemName, PricePerUnit, Qty, Date, CustId

3 | Add TotalAmount | orderdata | orderdata | TotalAmount = PricePerUnit * Qty

4 | Remove nulls/dupes | Both | Both | Exclude rows with nulls in key columns, remove duplicates

5 | Aggregate spend | ordersummary | customeraggregatespend | Group by Name/Date, sum TotalAmount

6 | SCD Type 2 update | ordersummary | ordersummary | Track historical changes

Appendix D: Business Rules

Rule ID | Area | Description | Source

BR-ETL-006 | Data Quality | Null/duplicate rows are to be excluded | Source files

BR-ETL-005 | Calculation | $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$ | Orders data

Appendix E: Process Flow

Step | Description | Dependency

1 | Load source data into raw tables | Source file availability

2 | Cleanse and transform data | Raw tables

3 | Enforce schema and add derived columns | Transformation logic

4 | Aggregate and load into target tables | Cleaned and structured data

5 | Apply SCD Type 2 logic on ordersummary | Up-to-date customer records